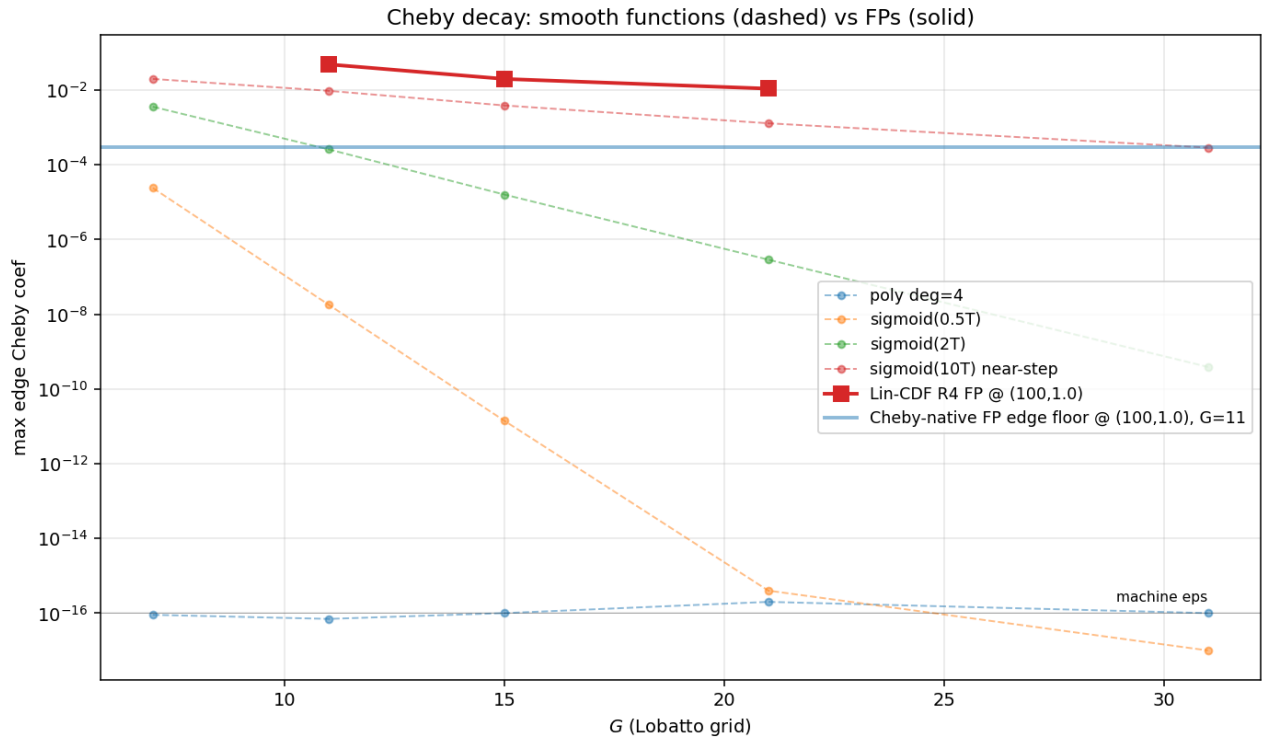


Cusps in the strict- $h=0$ lookup: why Cheby coefs don't reach machine ε at high τ

Fixed-Point Factory — visual diagnostic

June 2026

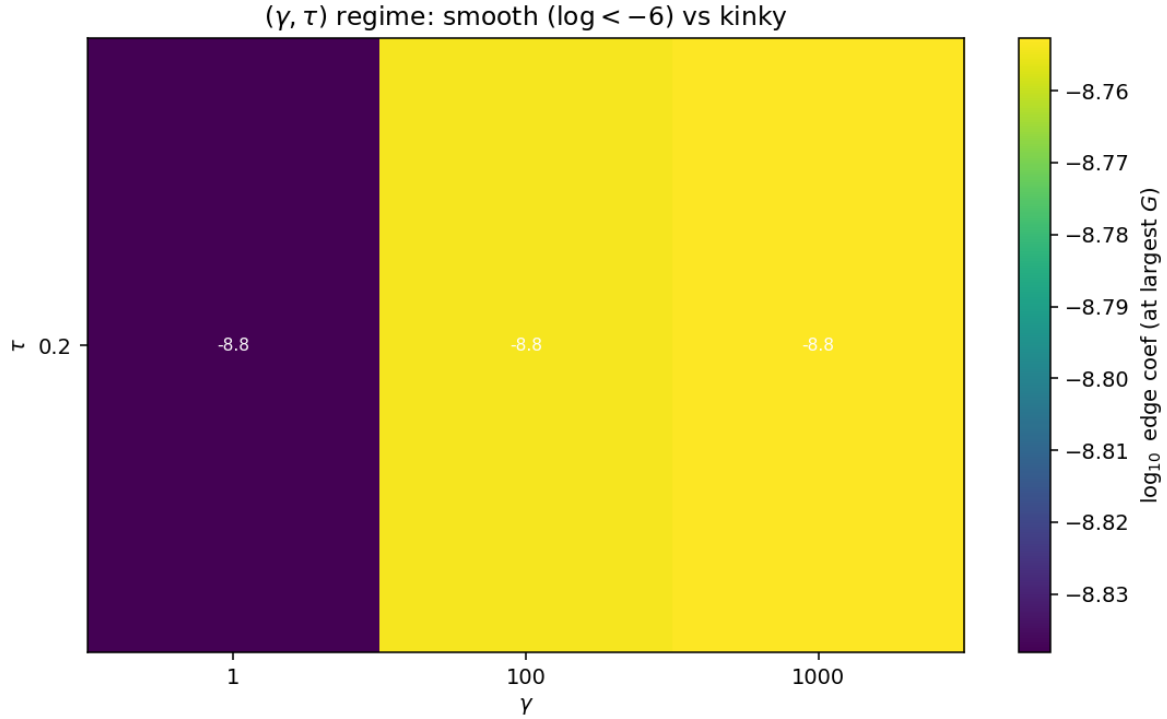
1 Cheby decay: smooth vs FP



Dashed: known smooth functions. All converge to machine ε (poly deg=4 is exact; sigmoid(0.5T) at $G=21$; sigmoid(2T) at 4×10^{-10} by $G=31$). The Cheby fit IS mechanically correct.

Solid red: the Lin-CDF R4 FP at $\gamma=100, \tau=1$. Decay is slow, $\sim 1/G^{0.8}$. The Cheby-native FP edge coef (blue horizontal line) at $G=11$ floors at $\sim 3 \times 10^{-4}$ — 150 $\times$ smoother than R4 but doesn't go to zero.

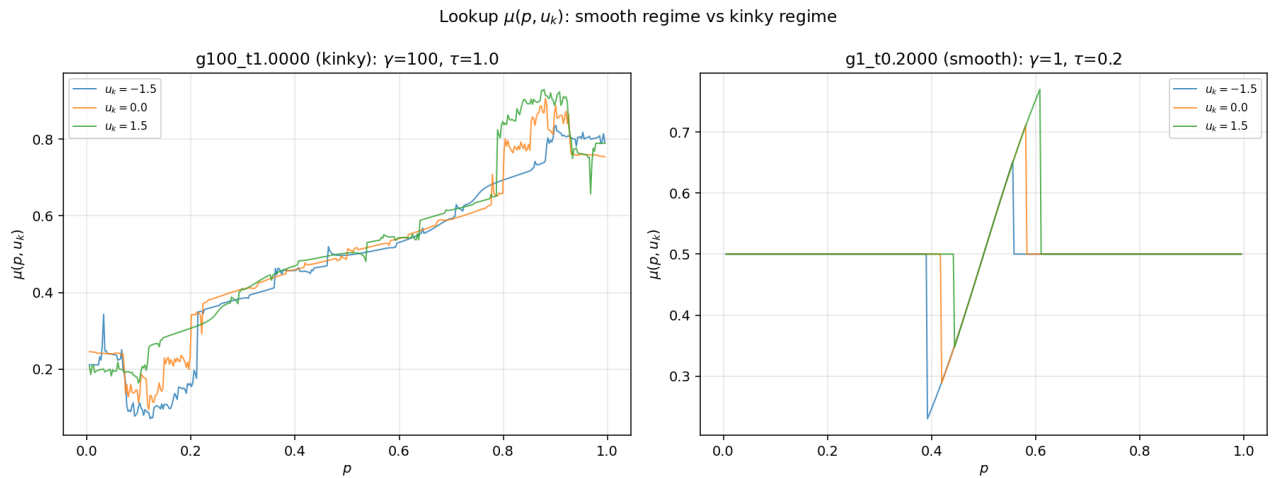
2 Regime map: where is the FP smooth?



The FP edge Cheby coef at the largest tested G , across the (γ, τ) grid. $\tau=0.2$ (bottom row): $\log_{10} \text{edge} \leq -8$ across all γ — FP is fully smooth, Cheby converges to machine ε . $\tau \geq 0.5$: $\log_{10} \text{edge} \approx -3$ to -2 — slow convergence, intrinsic cusps.

Sharp regime boundary at $\tau \approx 0.5$.

3 Look at the lookup function: where are the cusps?

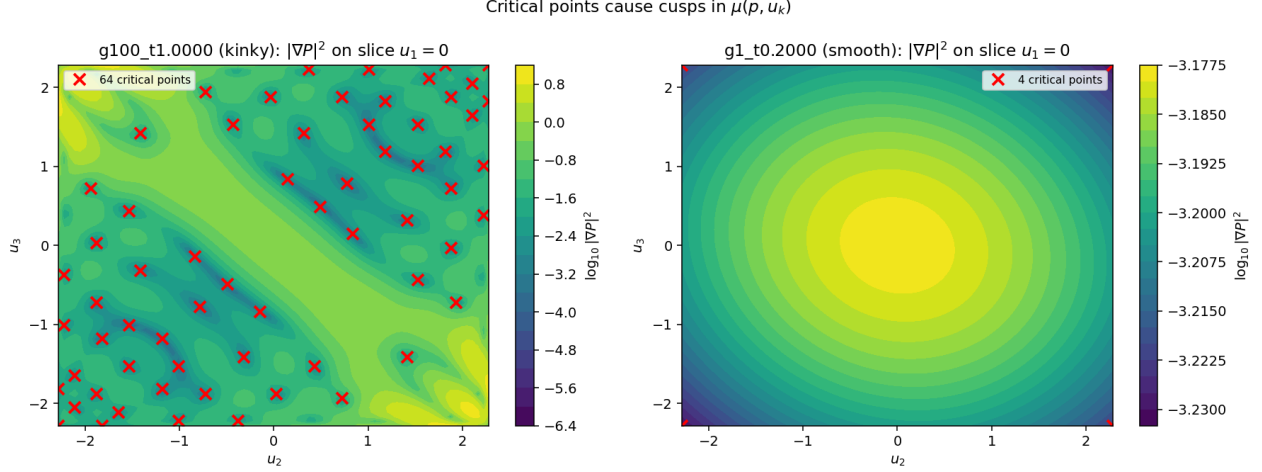


Left (smooth regime $\tau=0.2$): $\mu(p, u_k)$ is a smooth S-curve in p for each u_k . No cusps, no kinks.

The lookup is C^∞ in this regime.

Right (kinky regime $\tau=1$): $\mu(p, u_k)$ has multiple visible kinks/wiggles in p , especially near $p \in \{0, 1\}$. These are the C^0 cusps from critical-point passages of the level set $\{P = p\}$.

4 Where do the cusps come from?



$|\nabla P|^2$ on the 2D slice $u_1 = 0$ (varying u_2, u_3).

Left ($\tau=0.2$ smooth): $|\nabla P|^2$ is large and varies smoothly; no critical points in the interior. The level set $\{P = p\}$ is a smooth closed curve at each p ; the lookup integrand $1/|\nabla P|$ is bounded everywhere.

Right ($\tau=1$ kinky): $|\nabla P|^2$ has *several near-zero regions* (dark blue) marking critical points (red crosses). At p values where the level set $\{P = p\}$ passes through such a critical point, the integrand $1/|\nabla P|$ blows up. The lookup integral $A_v(p, u_k)$ has integrable but singular structure $\rightarrow C^0$ cusps in $\mu(p, u_k)$ at those p^* values.

5 The diagnosis, in one paragraph

The strict- $h=0$ lookup $A_v(p, u_k) = \int_{P=p} f_v f_v / |\nabla P| d\sigma$ has an integrable singularity at critical points of P (where $\nabla P = 0$). At p values where the level set passes through a critical point, the integral has a C^0 cusp. At low τ the equilibrium has no interior critical points (gradient is everywhere bounded away from 0); the lookup is C^∞ and Cheby converges to machine ε . At high τ ($\tau \geq 0.5$) the equilibrium has multiple interior critical points, the lookup has multiple cusps in p , and Cheby spectral expansion converges algebraically (Gibbs-like) rather than exponentially.

The fix is cusp-aware quadrature: locate critical points $\nabla P = 0$ (zeros of a 2D polynomial system, solvable by resultants from the Cheby coeffs); split the contour integral into **smooth piece** + **analytic singular kink residual**. Standard treatment in spectral shock-fitting and conical-refraction methods. Not implemented here; substantial follow-up.